

Laboratório III

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Inicialização

```
library(tidyverse)
library(leaflet)
```

Respostas

1.

```
airlines <- read_csv("../datasets/airlines.csv", show_col_types = FALSE)

airports <- read_csv(
  "../datasets/airports.csv",
  col_select = c("IATA_CODE", "CITY", "STATE", "LATITUDE", "LONGITUDE"),
  show_col_types = FALSE
)

process_chunk <- function(chunk, pos) {
  chunk |>
    drop_na() |>
    filter(!str_starts(DESTINATION_AIRPORT, "1")) |>
    group_by(DESTINATION_AIRPORT) |>
    summarize(
      chunk_delay = sum(ARRIVAL_DELAY),
      chunk_n = n(),
      .groups = "drop"
    )
}
```

```

chunks <- read_csv_chunked(
  file = "../datasets/flights.csv",
  callback = DataFrameCallback$new(process_chunk),
  chunk_size = 1000000,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- chunks |>
  group_by(DESTINATION_AIRPORT) |>
  summarize(
    mean_delay = sum(chunk_delay)/sum(chunk_n),
    .groups = "drop"
  )
flights |> head()

```

```

# A tibble: 6 x 2
  DESTINATION_AIRPORT mean_delay
  <chr>              <dbl>
1 ABE                5.80
2 ABI                4.08
3 ABQ                5.61
4 ABR               -3.39
5 ABY                8.68
6 ACK                4.91

```

2.

```

anti_join(flights, airports, by=c("DESTINATION_AIRPORT" = "IATA_CODE")) # Vazio. Não há aeroportos

```

```

# A tibble: 0 x 2
# i 2 variables: DESTINATION_AIRPORT <chr>, mean_delay <dbl>

```

```

flights <- left_join(flights, airports, by=c("DESTINATION_AIRPORT" = "IATA_CODE")) # Chaves
flights |> head()

```

```
# A tibble: 6 x 6
  DESTINATION_AIRPORT mean_delay CITY      STATE LATITUDE LONGITUDE
  <chr>              <dbl> <chr>    <chr>    <dbl>    <dbl>
1 ABE                5.80 Allentown PA       40.7     -75.4
2 ABI                4.08 Abilene   TX       32.4     -99.7
3 ABQ                5.61 Albuquerque NM       35.0    -107.
4 ABR               -3.39 Aberdeen  SD       45.4     -98.4
5 ABY                8.68 Albany    GA       31.5     -84.2
6 ACK                4.91 Nantucket MA       41.3     -70.1
```

3.

```
airports |>
  group_by(STATE) |>
  summarize(
    n_airport = n(),
    .groups = "drop"
  ) |>
  arrange(desc(n_airport))
```

```
# A tibble: 54 x 2
  STATE n_airport
  <chr>    <int>
1 TX         24
2 CA         22
3 AK         19
4 FL         17
5 MI         15
6 NY         14
7 CO         10
8 MN          8
9 MT          8
10 NC          8
# i 44 more rows
```

4.

```

this <- flights |>
  drop_na(LATITUDE, LONGITUDE) |>
  leaflet() |>
  addTiles() |>
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      DESTINATION_AIRPORT, " (", CITY, "-", STATE, ")<br>",
      "Mean arrival delay: ", round(mean_delay, 2), " min"
    )
  )
# this

```

O gráfico interativo só aparece na versão html. Tentar renderizá-lo como pdf gera um erro no processo.

Momento de compilação

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-08-29 12:38:22 -03"
```